

Kernza forage check

rolling report

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1 Trial specs

trial description	comparing forage yield of different wheatgrass lines
contact	TLI, spencer barriball@landinstitute.org
measurements	forage DM yield by plot at each forage harvest, field characterization, plant counts at establishment
trial start	2024 Sept
trial end	2026 Sept
row spacing	12"
quadrat dimensions	24" x 36" or 2 rows at 36" length = ~0.5m ²
quadrats per plot	1
quadrats per harvest	48
quadrats weighed for wet weight and dumped in field	0
quadrats back to campus for wet+dry weight, then discarded	0
quadrats back to campus for wet+dry in 95F, then ground for quality	48
management, forage	1 cutting at anthesis
management, fertility	fertilize @ 50 kg N ha once per year, spring or fall and as needed if crops look deficient
management, crop protection	herbicides (24D+dual), handweeding, gopher poisoning as needed
management, post harvest	cut to 3" and removed with new holland 36
irrigation	none
field cropping history	vetch, then soybean, soybean harvested in August right before planting
field preparation	soil was amended with P,K,S etc based on predicted removal from forage harvests over 4 years, rototilled, rolled, rototilled, planted and then immediately rolled post planting
field characterization	baseline soil samples were taken right after emergence plant counts confirmed successful iwg stand achieved (0-15cm, 4 pushes per block)

1.1 activities

Table 2: 2024 field activities

4/14	spring wheat planted bulk in r37
9/4	Fertilized one week prior. Rototilled, rolled, and royotilled again. The additional rototilling was because the seed bed was uneven after rolling. Planted plot length of 10.5 feet on setting 35. Standard IWG seeding depth. Rolled after planting.
9/26	Plant counts for new stands 2x 0.5meter lengths in different rows. Did not collect weed data
10/8	Applied 2, 4 d at ~1 pt per acre (1.3oz. per gallon plus 2oz. Class Act per gallon) to the experiment area plus around borders to control for broadleaf weeds using the 5 foot boom sprayer.
10/28	Fertilizer by hand by strips with urea to achieve 50 kg/ ha. (2 lbs per strip)
10/29	Soil sampled by block (did not get urea prills) for a baseline. 0-15 cm, 4 pushes per block.

Table 3: 2025 field activities

4/17	Fertilized with urea to achieve 60 lbs N per acre
5/19	Staging check
6/5	swapped metal falgs for plastic and moved them close to trial so they could mow alleys
6/30	Harvested all plots. Definitely at anthesis for most plots. 2x 36 inch lengths taken from middle two rows of plot. Katherine noted lodging of plots. Pulled all but two flags from field to allow for station to clear.
7/4	Chopped with new holland at some point. Actual date unknown- check with Darby and or Gary
8/14	Roughly marked the experiment boundary with tall flags and a whisker. Very hard to tell while the IWG is short and the area is weedy.

1.2 plot map

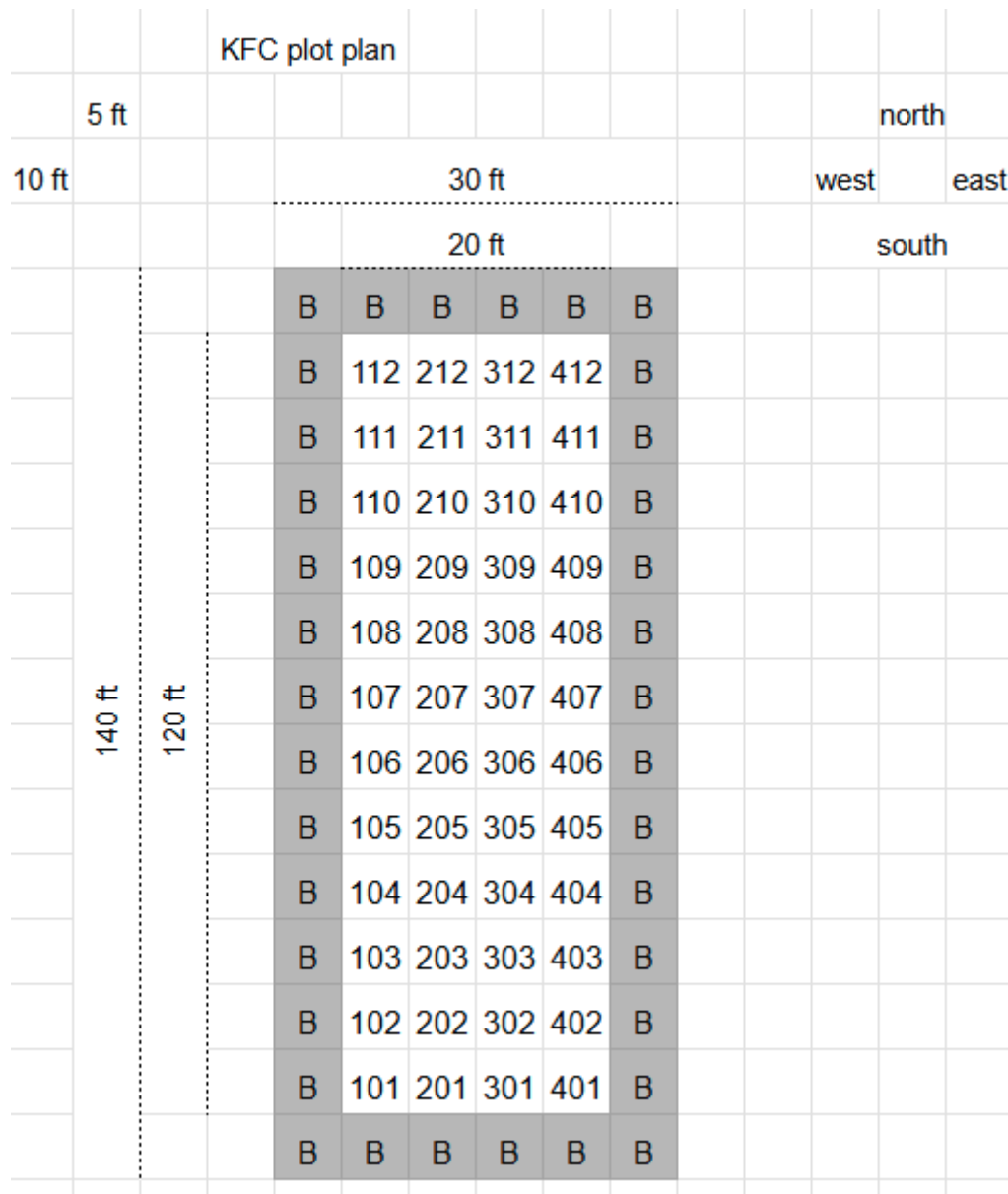


Figure 1: Plot map. Plots are 5 ft x 10 ft.

2 Population

Planted 4Sept2024

Plant counts collected 26Sept2024

Target population = 30 plants per meter

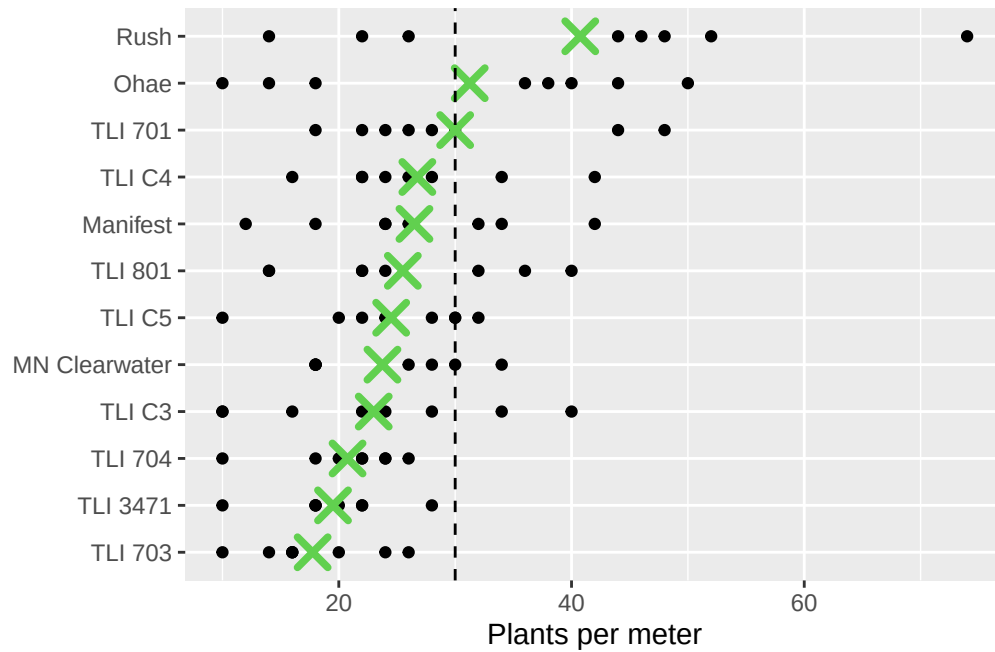


Figure 2: Plant counts 3 weeks after planting. X indicates mean. Dashed line indicates target population.

3 Yield

Harvested 30June2025 by hand using quadrats

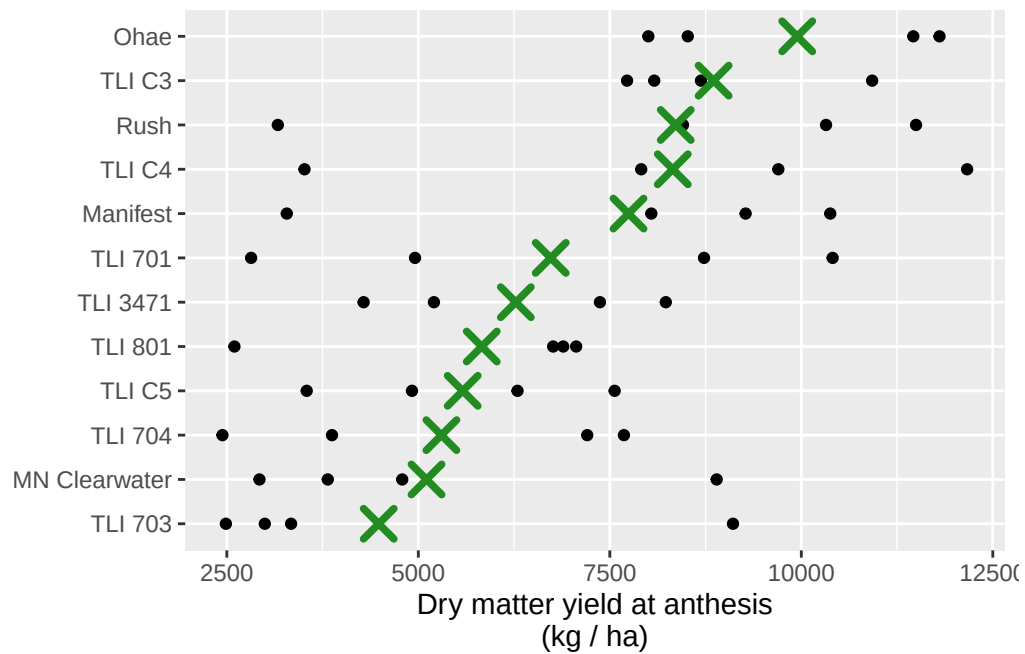


Figure 3: Forage yield at anthesis. X indicates mean.

Table 4: No differences among moisture content between varieties at harvest

Entry	Moisture content %
MN Clearwater	70.6
Manifest	72.0
Ohae	70.9
Rush	69.0
TLI 3471	71.7
TLI 701	67.9
TLI 703	74.8
TLI 704	70.9
TLI 801	74.7
TLI C3	74.0
TLI C4	67.5
TLI C5	75.7

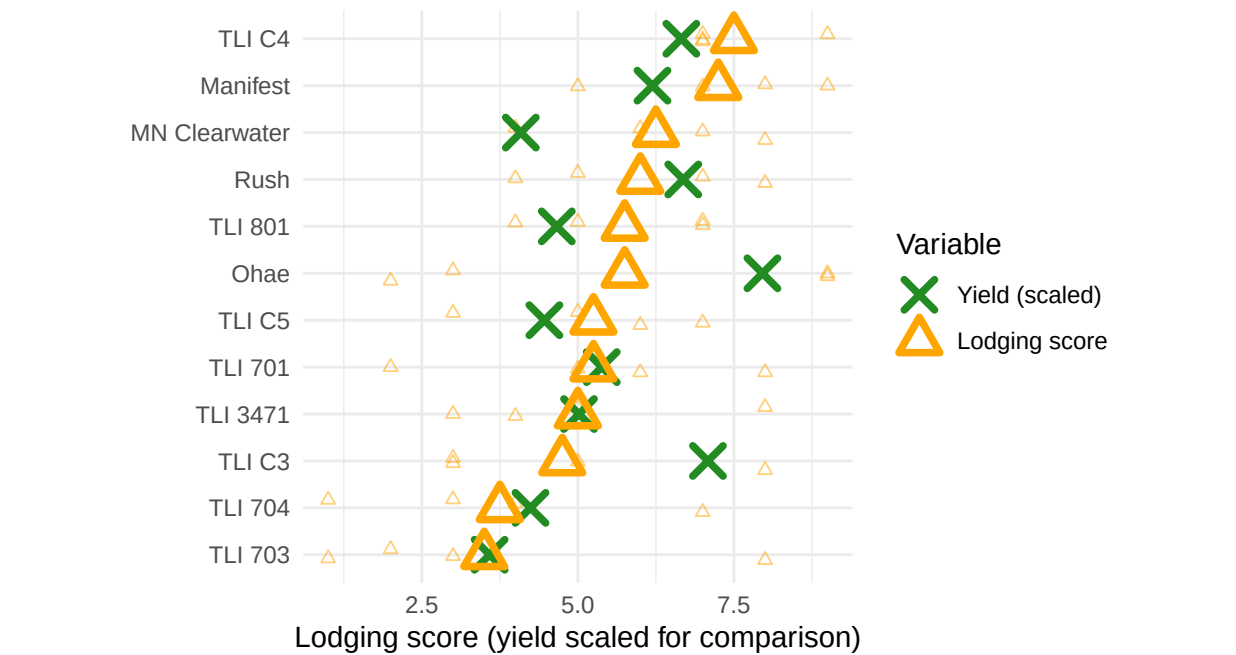


Figure 4: Forage lodging scores at harvest (anthesis). Yield presented for comparison. Notice some of the lower yielding plots had lower lodging as well.

4 Forage quality

To be determined...